5. Ensuring the input-output relation is

   (a) the responsibility of the algorithm and the right of the user.

   (b) the responsibility of the user and the right of the algorithm.

   (c) the responsibility of the algorithm but not the right of the user.

   (d) the responsibility of both the user and the algorithm.

6. If i = 5 before the assignment i := i-1 after the assignment, the value of i is

   (a) 5 (b) 4 (c) 3 (d) 2

7. If 0 < i before the assignment i := i-1 after the assignment, we can conclude that

   (a) 0 < i (b) 0 ≤ i (c) i = 0 (d) 0 ≥i

## SECTION - B

### Very Short Answers

1. Define an algorithm.
2. Distinguish between an algorithm and a process.
3. Initially,

   farmer, goat, grass, wolf = L, L, L, L

   and the farmer crosses the river with goat. Model the action with an assignment statement.
4. Specify a function to find the minimum of two numbers.
5. If $\sqrt{2}$ = 1.414, and the square_root() function returns -1.414, does it violate the following specification?

   **-- square_root (x)**

   **-- inputs: x is a real number , x ≥ 0**

   **-- outputs: y is a real number such that $y^2=x$**

## SECTION - C

### Short Answers

1. When do you say that a problem is algorithmic in nature?
2. What is the format of the specification of an algorithm?
3. What is abstraction?
4. How is state represented in algorithms?
5. What is the form and meaning of assignment statement?
6. What is the difference between assignment operator and equality operator?



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